ModelIt!

Student Activity Pack

Starlight and Distance: Mapping the Universe

Why the Brightest Star You See Might Not Be the Closest

Grade 05L2 | NGSS 5-ESS1-1 | Scale, Proportion, and Quantity

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

9 Post-Assessment 11

1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Look at the stars tonight (or imagine them). Are the brightest ones the closest? How do you know?**

2. **If you hold a flashlight close to your face and then far away, what changes? Does the flashlight change?**

3. **How do you think scientists figure out how far away a star is?**

4. **What is a light-year? Take your best guess.**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Star Actual Brightness | Distance from Earth | Apparent Brightness | Light Travel Time | Observable Detail

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Nearby Star

Moderate brightness, very close

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Distant Giant

Maximum brightness, very far away

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Side-by-Side

Dim close star vs bright distant star

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Telescope Power — A stronger telescope can resolve more detail at greater distances

Star Color — A star's color tells us its temperature and actual brightness

Atmospheric Interference — Earth's atmosphere blurs starlight, reducing observable detail

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How do we know how far away stars are if we can't travel to them? Use evidence from your simulation to support your answers.

1. **Why is it misleading to say 'that star is the brightest in the sky'?**

2. **When you look at a star 100 light-years away, what are you actually seeing?**

3. **How did this model change your understanding of what you see in the night sky?**

4. **Why is it important for astronomers to know actual brightness, not just apparent brightness?**

5. **What limitations does this model have? What would make it more realistic?**

8 STEM Challenge: Design a Star Distance Calculator

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YOUR ENGINEERING MISSION

Design a method to determine how far away an unknown star is using only its apparent and actual brightness.

THE SCENARIO

NASA has discovered a new star. They know its actual brightness from its color and size, and they can measure its apparent brightness from Earth. Your team must design a method to calculate its distance.

Engineering Design Thinking

1. WHAT MEASUREMENTS WILL YOU NEED (ACTUAL BRIGHTNESS, APPARENT BRIGHTNESS)?

2. WHAT IS THE RELATIONSHIP BETWEEN DISTANCE AND APPARENT BRIGHTNESS?

3. HOW WILL YOU PRESENT YOUR METHOD SO OTHER SCIENTISTS CAN USE IT?

4. HOW ACCURATE DO YOU THINK YOUR METHOD IS? WHAT COULD CAUSE ERRORS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Astrophysicists and Stellar Astronomers study stars to understand the universe. They use the exact relationship you modeled today — comparing actual and apparent brightness — to map the cosmos. They earn $80,000-$150,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 In the starlight model, a star with high Actual Brightness is placed at 90% Distance from Earth. What does the model predict for Apparent Brightness?

- **A.** Apparent Brightness is very high because the star is very bright
- **B.** Apparent Brightness is low because distance greatly reduces how bright the star looks from Earth
- **C.** Apparent Brightness equals Actual Brightness because distance does not matter
- **D.** Apparent Brightness is zero because distant stars are invisible

2 Star A has low Actual Brightness but is very close to Earth. Star B has very high Actual Brightness but is very far away. Which star looks brighter from Earth?

- **A.** Star B always looks brighter because it produces more light
- **B.** They always look the same because brightness and distance cancel out
- **C.** Star A could look brighter than Star B because proximity can outweigh luminosity
- **D.** Neither star is visible from Earth

3 What type of relationship exists between Distance from Earth and Apparent Brightness?

- **A.** Positive: farther stars look brighter
- **B.** No relationship: distance does not affect how bright stars appear
- **C.** Negative: greater distance means the star looks dimmer from Earth
- **D.** Positive: distance magnifies starlight like a lens

4 The sun appears brighter than Betelgeuse from Earth, even though Betelgeuse is about 100,000 times more luminous. What does the model tell us about why?

- **A.** Betelgeuse's light is blocked by dust clouds
- **B.** The sun is 93 million miles away while Betelgeuse is about 700 light-years away, and distance dramatically reduces apparent brightness
- **C.** The sun is actually more powerful than Betelgeuse
- **D.** Betelgeuse is not a real star

5 When we look at a star that is 500 light-years away, what are we actually seeing?

- **A.** The star as it is right now, in real time
- **B.** The star as it was 500 years ago, because its light took 500 years to reach us
- **C.** A prediction of what the star will look like in 500 years
- **D.** An illusion because stars are too far away to see